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The Global Dataset of Cyber Incidents Data Handbook

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European **Repo**sitory of **C**yber Incidents

Thank you for your interest in our cyber incident dataset.

We are releasing the Global Dataset of Cyber Incidents in Version 1.3 as an extract of our backend database. This official release contains fully consolidated cyber incident data reviewed by our interdisciplinary experts in the fields of politics, law and technology across all 60 variables covered by the Repository. Version 1.3. covers the years 2000 – 2024 entirely. The Global Dataset is meant for reliable, evidence-based analysis. If you require real-time data, please refer to the download option in our TableView or contact us for special requirements (including API access).

The dataset is offered in the form of Excel- and CSV-files and has been simplified for this release. Our backend-database allows us to model very complex aspects of each cyber incident which cannot easily be transferred into a single table. Please note that this has some implications for the given dataset. The Guardrails here are meant to support you in your analysis.

The global dataset contains exactly one incident in each row, identified by incident id and a name. Since many categories allow for multiple codes, they are stored in each column separated by a semicolon. Refer to the codebook and handle data accordingly where it states „multiple codings possible".

1. Missings

The dataset uses three different types of missing values.

1. Not available: This is the default missing value for categories not coded/not possible to code in the specific instance.
2. Unknown: This means that the information is empirically not known. The best example for this is the initiator country. Since, e.g., transnational criminal networks sometimes cannot be attributed to specific states, we may know the initiator name but not their country of origin.
3. Not attributed: This is the missing value for attributions. Because all attribution fields are related with each other, this signals that there is no attribution at all for the incident, which means there is no information about the initiator available.
4. (empty): Remember that our default value „Not available" will not be present for missings created when you e.g., split the multi-codes for creating a long format. In this case you need to handle empty fields depending on your software.

1. Operation type and Incident type

For the type of each incident, we are using a modular system that allows us to code multiple types for one single incident. This allows for much greater flexibility but commonly creates some confusion. For example, Ransomware incidents, we would code as *„Hijacking with misuse"* (because malware has been used), *„Disruption"* (because the encrypted data is unavailable to the target) and *„Ransomware"* (because it is a specific type). If there is indication for a double extortion scheme, we would add „Data theft" if data was stolen, or *„Data theft & doxing"* if there is evidence that the data has already been leaked.

In order to make this more accessible for less specific analysis, we aggregated the incident types into the „operation type" column which contains only one type per incident with the categories *"DDoS/Defacement", "Data theft", "Hijacking without Misuse", "Wiper", "Hack and leak", "Hijacking with Misuse", "Ransomware", "Other", and "Not available".* Note that this ignores the empirical complexity of e.g., the double extortion scheme.

2. Separate/Nested Entities

Certain fields are stored as separate entities in our database, allowing us to better model the relationships between them. For example, we know for each specific receiver country which exact entities in what sectors have been affected. This is currently not directly modelled in the Global Dataset extract as these fields are, like multi-code fields, only separated by semicolons. While you could try to retrieve relationships from the order, we recommend against this because there are more deeply nested fields. To solve this issue, we provide special datasets for Receivers and Initiators, as well as for dyadic relationships between states.

Table 1: Deeply nested entities – overview

Entity	Field	Subfield_1	Subfield_2
Receiver*	receiver_name		
	receiver_country		
	receiver_category*	receiver_subcategory*	
Attribution*	attribution_basis		
	attribution_type		
	attributing_country*		
	attributing_actor*		
	attribution_date		
	Initiator*	initiator_name	
		initiator_category*	initiator_subcategory
		Initiator_countries*	
	legal_attribution_reference		
Political Response*	political_response_ responding_country*		
	political_response_date		
	political_response_actor*		
	political_response_type*	political_response_subtype	
Legal Response	legal_response_respondin g_country*		
	legal_response_date		
	legal_response_actor*		
	legal_response_type*	legal_response_subtype	

* multiple codings possible

Code and Subcode fields are always related 1:1 which means the first entry for the subcode is a child of the first one; and so on. In the global dataset, this will lead to duplicated codes if there are multiple subcodes coded for one code, multiple sectors (receiver_subcategory) affected within critical infrastructures (receiver_category) will have a duplicate critical infrastructure code for each sector.

For many analysis cases, you can safely remove these duplicates. In general, we recommend summing up your counts based on unique values of the incident ID, assuming the number of incidents is of interest.

2. Receivers

Our database stores receivers as separate entities which cannot be reproduced in the aggregated Excel-file. The relationship issue holds for all receiver_* fields (see Table 1). For this reason, we offer the additional Receiver Dataset in this package, which comes in the „long“ format, displaying the relationships between receiver name, country, category and subcategory for each unique receiver and incident.

If you are merely interested in aggregated data, e.g., what sectors have been affected by an incident and not in a specific country, you may want to disaggregate the receiver_subcategory column by semicolon and remove duplicates (duplicates coming from e.g., the same sector being affected in multiple countries).

Note on Regions: Since a country can be part of multiple regions, regions are delivered as a list for each receiver country. Alternatively, you could assign the region based on the respective receiver countries manually. In this case, remember that Britain left the European Union in 2021. We are automatically considering this in our region codings.

3. Attributions & Initiators

Similar to receivers, our database stores attributions as separate entities. This means that there is a link between all attribution_* fields. We offer the disaggregated Attribution Dataset to make these relations more accessible. Attributions are a critical problem in cyber conflict research, and we are very meticulous about representing them properly. This means in our dataset, there may be a lot of different attributions for a single incident, and it is totally possible that these attributions do not share the same amount of precision or even contradict each other (e.g., they disagree on the state-affiliation of an initiator or even disagree on the initiator country).

If you are merely interested if an incident has been attributed or how often, refer to the number_attributions column. Note that it is a valid coding when certain Attribution fields are not available e.g., when the initiator country is not available, but the initiator name is coded, the country of origin for this initiator is not known (often the case for transnational collectives or criminals).

You may notice that there are multiple attribution rows with the same attribution ID in the Attribution Dataset. This is the case when joint attributions occur, where multiple actors or multiple states make a joint attribution statement which means they share the exact same attribution as opposed to making individual statements (but may or may not refer to the same threat actor).

Note that the Repository does not make attribution statements. Only if there is a reported Attribution, we store this information. This also means we do not code initiators by our own virtue but refer to one of our stored attributions as „settled attribution". This may be the only attribution available, or the most widely shared attribution. The settled attribution can be inferred from the respective column in the dataset.

To make the dataset more accessible, the global dataset in contrast to the attribution dataset only contains the settled attribution. However, there can still be multiple initiators if the settled attribution refers to multiple ones.

If you plan a comprehensive analysis of attributions, do not hesitate to contact us, so we can sort out the little perks of the attribution coding together.

4. Legal & Political Responses

Legal and Political responses are also separate entities. We currently do not offer respective extracts. In the case that there are multiple responses, they are (like other fields) separated by semicolons and the order is retained across the response-columns.

5. Dates

For dates including start & end date, attribution date and political/legal response date, there is not always sufficient information to code the information for the specific day. In some cases, we may only know the year or the month. In the aggregated dataset, we set this to the first of month or the first of January respectively, to make analysis easier. In the Attribution Dataset, we show the specific Year/Month/Day available as individual columns.

This can be important if you work with a resolution higher than yearly aggregated data, as there may be spikes at the start of a year/month. If you require a higher resolution, please don't hesitate to contact us. The information on the resolution of dates is stored in the backend-database but currently isn't exported for technical reasons.

6. Sources

We are transparent about every source we have on each incident. Like all other columns, the sources are contained as URLs in the source_url column. Note that in order to retain Excel-compatibility, there may be single spaces behind each semicolon in the Excel extract.

7. How to approach common analysis cases

Many questions may focus on numbers of cyber incidents in various configurations. For many such questions, you can roughly build your work on the following example. Here, we will use the number of incidents for a specific country:

1. Disaggregate the column of interest by splitting the receiver_country column on the semicolon; that is, create an additional row for each unique receiver country in the row and column.
2. Aggregate the number of unique cyber incidents for each country based on the incident_id. Depending on your analysis software, this includes grouping on country and counting the number of unique incident_id.
3. Now you receive the number of incidents targeting the specific country. Note that the sum of all incidents targeting all countries may be higher than the total sum of incidents, because incidents can target multiple countries.

The Dyadic Dataset

With version 1.3 we are releasing the EuRepoC Dyadic Dataset. This builds on our backend-database to offer you a view on our data that may be more intuitive for answering the question

who did what to whom, and how bad was it?

The dyadic dataset puts state dyads in the focus. Each row in the dataset represents one cyber incident in a specific dyad. Because incidents may affect multiple receivers, single incidents can be duplicated in this format, when they affected multiple countries.

It is important to note that the countries are countries of origin and need to be viewed in context of initiator_type. Even if an initiator_country is coded, it may not be an incident caused by state-actors but by non-state actors with origin in the respective country. Initiator and receiver can also involve

1. Missing values (e.g., Not attributed)
2. Regions, e.g., it is possible for an incident to target a whole region.

The dyadic dataset simplifies some values.

1. Receiver names, categories, subcategories still come as a list, but duplicates are removed, and values depend on the receiver countries. For example, if the receiver_country of a row is the United States, receiver_categories lists all categories affected in the United States.
2. Attribution complexity is omitted and only the primary settled initiator is shown.
3. There are dummy variables for the incident type to better represent the combinations for each incident.